CHE 321 HW # 7 Cesearo Dilosa

P8-4A.

① $A \rightarrow X \qquad r_{1X} = K_1 C_A^{1/2} \qquad K_1 = 0.004$

② $A \rightarrow B \qquad r_{2B} = K_2 C_A \qquad K_2 = 0.3$

③ $A \rightarrow Y \qquad r_{3Y} = K_3 C_A^2 \qquad K_3 = 0.25$

* Assume reactor is PFR

a.

- $S_{B/X} = \dfrac{r_B}{r_X} = \dfrac{K_2 C_A}{K_1 C_A^{1/2}}$

- $S_{B/Y} = \dfrac{r_B}{r_Y} = \dfrac{K_2 C_A}{K_3 C_A^2}$

- $S_{B/XY} = \dfrac{r_B}{r_X + r_Y} = \dfrac{K_2 C_A}{K_1 C_A^{1/2} + K_3 C_A^2}$

- $C_A = C_{T0} \dfrac{F_A}{F_T} \cancel{\dfrac{P}{P_0}} \cancel{\dfrac{T_0}{T}}$

* Assume isothermal and no pressure drop

$\hookrightarrow C_{T0} = \dfrac{P}{RT} = \dfrac{4 \text{ atm}}{0.08206 \frac{\text{atm}\cdot\text{L}}{\text{K}\cdot\text{mol}} \times 300\text{ K}}$

$C_{T0} = 0.162 \ \frac{\text{mol}}{\text{dm}^3}$

$C_{T0} = C_{A0}$

- $\dfrac{dF_A}{dV} = r_A = -(r_{1X} + r_{2B} + r_{3Y})$

$\hookrightarrow r_{1A} = -r_{1X}$

$\hookrightarrow r_{2A} = -r_{2B}$

$\hookrightarrow r_{3A} = -r_{3Y}$

- $\dfrac{dF_B}{dV} = r_B$ • $\dfrac{dF_X}{dV} = r_X$ • $\dfrac{dF}{dV} = r_Y$

- $\dfrac{d(S_{B/XY})}{dC_A} = 0 = \dfrac{0.5 K_2 K_1 C_A^{*1/2} - K_2 K_3 C_A^{*2}}{(K_1 C_A^{*1/2} + K_3 C_A^{*2})^2}$

* Used Mathematica

$C_A^* = \left(\dfrac{K_1}{2K_3}\right)^{2/3} = 0.04 \ \frac{\text{mol}}{\text{dm}^3}$

This is the C_A used in a CSTR to maximize $S_{B/XY}$.

c.) $C_i = C_{T0} \frac{F_i}{F_T} \frac{\cancel{P}}{\cancel{P_0}}\overset{1}{} \frac{\cancel{T_0}}{\cancel{T}}\overset{1}{}$

↳ for A, B, X, and Y

↳ $C_{T0} = 0.162 \frac{mol}{dm^3}$

↳ $F_T = F_A + F_B + F_X + F_Y$

d.) $X_A = \frac{C_{A0} - C_A^*}{C_{A0}} = 0.753$

b.) Using a CSTR since the reactor can operate at a constant C_A. (C_A^* will be used)

$$V = \frac{v_0 (C_{A0} - C_A^*)}{-r_A^*} = \frac{\overset{10\ dm^3/min}{v_0} \left(\overset{0.162}{C_{A0}} - \overset{0.04 \frac{mol}{dm^3}}{C_A^*} \right)}{-\left(\underset{0.004}{k_1} C_A^{*\,1/2} + \underset{0.3}{k_2} C_A^* + \underset{0.25}{k_3} C_A^{*2} \right)}$$

$$= 92.4\ dm^3$$

c.) • $V = \frac{v_0 C_B}{-r_B} = \frac{v_0 C_B}{k_2 C_A} \Rightarrow C_B^* = \frac{V^* k_2 C_A^*}{v_0}$

Because first reactor is operated at C_A^*

$$= \frac{92.4\ dm^3 \times 0.3 \frac{1}{min} \times 0.04 \frac{mol}{dm^3}}{10\ dm^3/min}$$

$$= 0.11 \frac{mol}{dm^3}$$

• $C_X^* = \frac{V^* k_1 C_A^{*\,1/2}}{v_0} = 0.007 \frac{mol}{dm^3}$

• $C_Y^* = \frac{V^* k_3 C_A^{*2}}{v_0} = 0.004 \frac{mol}{dm^3}$

(e.) To achieve 99% conversion a PFR should follow the first CSTR

$$\frac{dX}{dV} = \frac{-r_A}{F_{A0}} \Rightarrow V_{PFR} = C_{A0} v_0 \int_{0.753}^{0.99} \frac{dX}{K_1 C_A^{1/2} + K_2 C_A + K_3 C_A^2}$$

$\hookrightarrow C_A = C_{A0}(1-X)$

$$V_{PFR} = 94.70 \text{ dm}^3$$

* I used Mathematica to solve the integral.

(f.) • $V = \dfrac{\cancel{F_{B0}}_0 - F_B}{-r_B} = \dfrac{C_B v_0}{K_2 C_A} \Rightarrow C_B = \tau K_2 C_A$

$\hookrightarrow K_2(T) = K_2(300K) \exp\left[\frac{E}{R}\left(\frac{1}{300K} - \frac{1}{T}\right)\right]$

* C_B can be plotted as a function of temp. I would recommend whatever temp gives a maximum value for C_B.

(g.) Since $P_A = C_A RT$, I would choose P_A^* $(= C_A^* RT)$ because this would at least give high $S_{B/XY}$ which is a good starting point.

(P8-10B.) *Used Matlab with the help of Claude (AI) to help with syntax and labeling.